Lecture 1: Introduction to medical imaging

Imaging volume and complexity are increasing

RADIOLOGY BUSINESS

FOR LEADERS NAVIGATING VALUE-BASED CARE

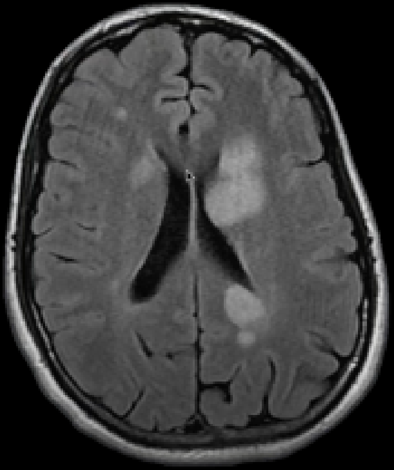
Feeling overworked? Rise in CT, MRI images adds to radiologist workload

Michael Walter

Both CT and MRI utilization and the number of images being collected for those exams have increased significantly in recent years, according to recent study published in *Academic Radiology*. The authors believe this could potentially lead to an increase in errors made by overworked radiologists.

Mayo did hire additional radiologists to assist with this increase in utilization and images, but McDonald and colleagues found that it wasn't enough. In 1999, a radiologist interpreting CT scans was required to interpret 2.8 images per minute. In 2010, that same number was over 19 images per minute. Likewise, a radiologist interpreting MRI scans was required to interpret 3 images per minute in 1999, but that number jumped to almost 12 per minute in 2010.

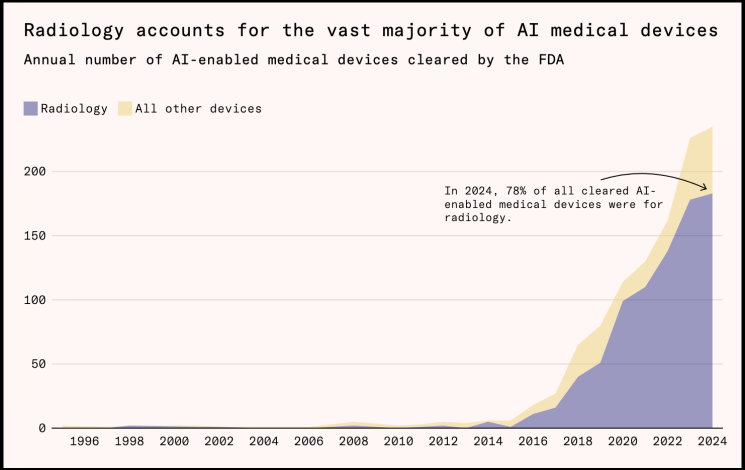
Medical Knowledge is highly subspecialized



Glioma?
ADEM?
HIV encephalopathy?

Trends of AI in Radiology

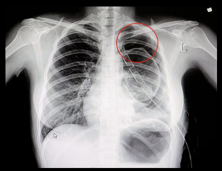
- Geoffrey Hinton's statement on radiology and AI in 2016: "It's just obvious that within five years deep learning is going to do better than radiologists."
- A significant rise in AI-enabled medical devices cleared by FDA.



Source: Food and Drug Administration (2025)

Primary Modalities in Medical Imaging

2D PROJECTION



Source: Livhospital

X-Ray

- High-energy EM radiation
- Density for attenuation
- Best for: Fractures, Chest

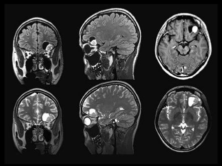
3D VOLUME

Source: Radiopaedia

CT

- Rotating source + detector
- Hounsfield Unit (HU) scale
- Best for: Trauma, Tumors

3D SOFT TISSUE

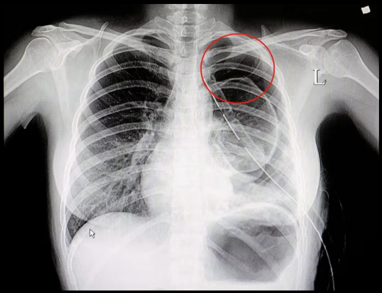


Source: AppliedRadiology

MRI

- Hydrogen proton
- Soft-tissue contrast
- Best for: Brain, Ligaments

Medical Imaging: X-Ray

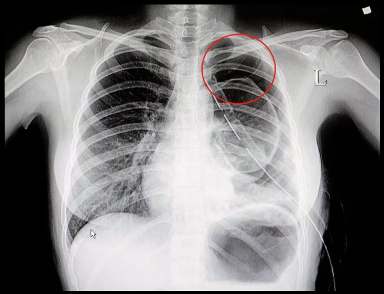


2D Projection View

Medical Imaging: X-Ray

Imaging Mechanism

High-energy EM radiation is transmitted through the body.



2D Projection View

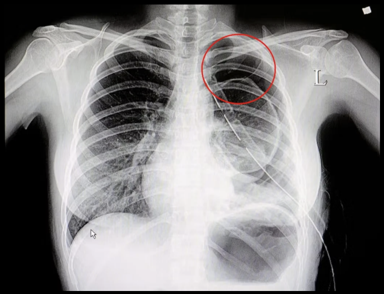
Medical Imaging: X-Ray

Imaging Mechanism

High-energy EM radiation is transmitted through the body.

Technical Features

- Attenuation.
- Projection.



2D Projection View

Medical Imaging: X-Ray

Imaging Mechanism

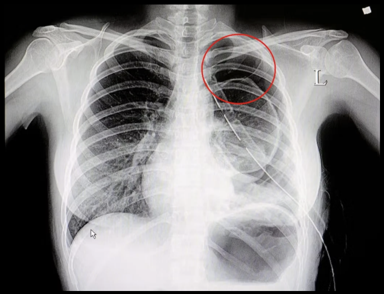
High-energy EM radiation is transmitted through the body.

Technical Features

- Attenuation.
- Projection.

Clinical Applications

Bone fractures, Dental, Chest screening.



2D Projection View

Medical Imaging: Computed Tomography (CT)

Cross-sectional & 3D View

Medical Imaging: Computed Tomography (CT)

Imaging Mechanism

Rotating X-ray source + detectors reconstruct cross-sectional slices.

Cross-sectional & 3D View

Medical Imaging: Computed Tomography (CT)

Imaging Mechanism

Rotating X-ray source + detectors reconstruct cross-sectional slices.

Technical Features

- Hounsfield Units (HU).
- Tomographic Reconstruction.

Cross-sectional & 3D View

Medical Imaging: Computed Tomography (CT)

Imaging Mechanism

Rotating X-ray source + detectors reconstruct cross-sectional slices.

Technical Features

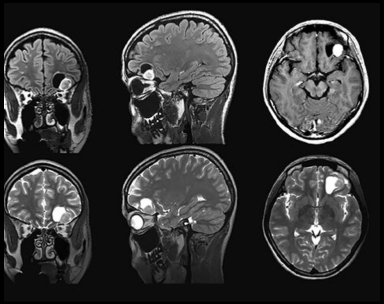
- Hounsfield Units (HU).
- Tomographic Reconstruction.

Clinical Applications

Complex fractures, trauma assessment, internal bleeding, tumor detection and staging.

Cross-sectional & 3D View

Medical Imaging: MRI

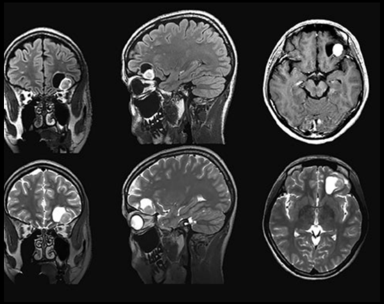


High Soft-Tissue Contrast

Medical Imaging: MRI

Imaging Mechanism

Strong magnetic fields + RF pulses excite hydrogen nuclei; signals are reconstructed into images.



High Soft-Tissue Contrast

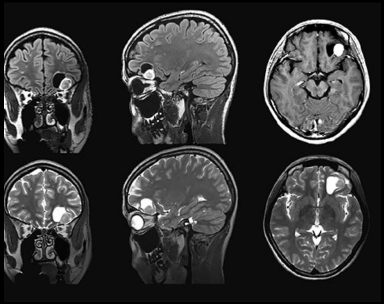
Medical Imaging: MRI

Imaging Mechanism

Strong magnetic fields + RF pulses excite hydrogen nuclei; signals are reconstructed into images.

Technical Features

- No Ionizing Radiation.
- Soft-Tissue Contrast.
- T1 vs T2.



High Soft-Tissue Contrast

Medical Imaging: MRI

Imaging Mechanism

Strong magnetic fields + RF pulses excite hydrogen nuclei; signals are reconstructed into images.

Technical Features

- No Ionizing Radiation.
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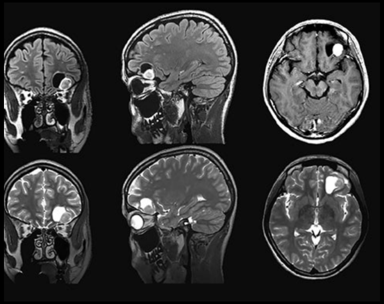
- T1 vs T2.

Clinical Applications

Brain and spinal cord imaging, ligament and tendon injuries, pelvic and abdominal soft tissues.



High Soft-Tissue Contrast